

Dialogue Systems Evaluation

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آکادمی‌ها



Evaluating Chatbots and Task-based Dialogue

➤ Task-based dialogue:

- ▶ Mainly by measuring task performance

➤ Conversational chatbots:

- ▶ Mainly by human evaluation



➤ **Evaluation of Conversational Chatbots**

➤ **Evaluation of Task-based Dialogue**



Chatbots are evaluated by humans

- **Participant evaluation:** The human who talked to the chatbot assigns a score
- **Observer evaluation:** third party who reads a transcript of a human/chatbot conversation assigns a score



Participant Evaluation

➤ Human chats with model for 6 turns and rates 8 dimensions of quality:

- ▶ avoiding repetition
- ▶ interestingness
- ▶ making sense
- ▶ fluency
- ▶ listening
- ▶ inquisitiveness
- ▶ humanness
- ▶ engagingness



Participant Evaluation

➤ **Avoiding Repetition:** How repetitive was this user?

- ▶ Repeated themselves over and over
- ▶ Sometimes said the same thing twice
- ▶ Always said something new



Participant Evaluation

▶ **Making sense:** How often did this user say something which didn't make sense?

- ▶ Never made any sense
- ▶ Most responses didn't make sense
- ▶ Some responses didn't make sense
- ▶ Everything made perfect sense



Participant Evaluation

► **Engagingness:** How much did you enjoy talking to this user?

- Not at all
- A little
- Somewhat
- A lot



Observer Evaluation: acute-eval

➤ Annotators look at two conversations (A + B) and decide which is better:

➤ **Engagingness:**

▶ Who would you prefer to talk to for a long conversation?

➤ **Interestingness:**

▶ If you had to say one of these speakers is interesting and one is boring, who would you say is more interesting?

➤ **Humanness:**

▶ Which speaker sounds more human?

➤ **Knowledgeable:**

▶ If you had to say that one speaker is more knowledgeable and one is more ignorant, who is more knowledgeable?



Observer Evaluation: acute-eval

➤ The ACUTE-EVAL method (Li et al., 2019)

Hi how are you today	Hello there, how are you?
Hi, I'm pretty good! Just listening to some aerosmith, they're my fave :) whatre you up to?	I am doing great. How are you?
I am listening to some italian music	I am great, I did something crazy for me and colored my hair blue!
Italian music, nice! What do you do for work?	I have a daughter and a son who also love blue colored balls. You should meet them
I'm a professional wrestler	Well that neat, I got a new car my mother gave so maybe I could see them!
Oh wow! I'm a construction worker, I've been doing it since I was 16.	It is a beautiful city. And, I try to be... Just cannot afford a bigger house atm.
What do you do for a living	I am sorry to hear that, I feel bad going out of town for spring break now.
I'm a construction worker haha, just told you! Whats your favorite type of food?	Ok. I going to school in the spring for casino manager
My favorite is garlic bread	Well I turn 29 next week, I wonder if that is a good age to apply as one.
Yum! I love mexican food, so good :) do you have any hobbies?	My grandmother just died from lung cancer, sucks
I like to workout and go to the gym	
We're a bit different- I love watching nascar and ufc. They're so fun!	
Who would you prefer to talk to for a long conversation?	
☐ I would prefer to talk to Speaker 1 ☐ I would prefer to talk to Speaker 2	
Please provide a brief justification for your choice (a few words or a sentence)	
Please enter here...	



Automatic Evaluation

- Automatic evaluation metrics are used for comparing machine generated text with gold texts, like
 - ▶ BLEU (mainly used for machine translation)
 - ▶ ROUGE (mainly used for text summarization)
- Automatic evaluation metrics are generally not used for chatbots
 - ▶ They are so many possible responses for a given turn
 - ▶ They correlate poorly with human judgements



Adversarial Evaluation

- ▶ One current research direction toward automatic evaluation:
Adversarial Evaluation
 - ▶ Inspired by the Turing Test
 - ▶ Train a ``Turing-like" classifier to distinguish between human responses and machine responses
 - ▶ The more successful a dialogue system is at fooling the evaluator, the better the system



➤ Evaluation of Conversational Chatbots

➤ **Evaluation of Task-based Dialogue**



Task-based Evaluation

- End-to-end evaluation
 - ▶ Measure absolute task success
 - ▶ Measure user satisfaction
- Slot Error Rate for a Sentence



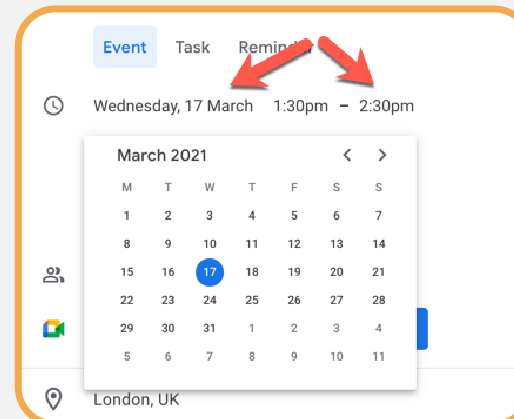
Evaluating by Task Success!

➤ Measure absolute task success

▶ Did the system book the right plane flight?



▶ Did the system put the right event on the calendar?





Evaluating by User Satisfaction

➤ More Fine-grained Metrics

➤ A User Satisfaction Survey:

TTS Performance	Was the system easy to understand ?
ASR Performance	Did the system understand what you said?
Task Ease	Was it easy to find the message/flight/train you wanted?
Interaction Pace	Was the pace of interaction with the system appropriate?
User Expertise	Did you know what you could say at each point?
System Response	How often was the system sluggish and slow to reply to you?
Expected Behavior	Did the system work the way you expected it to?
Future Use	Do you think you'd use the system in the future?



Evaluating Slot Filling

➤ Slot Error Rate for a Sentence

of inserted/deleted/substituted slots

of total reference slots for sentence

➤ Precision, Recall, F-score for slot filling evaluation



Evaluation Metrics: Slot Error Rate

"Make an appointment with Chris at 10:30 in Gates 104"

Slot	Filler
PERSON	Chris
TIME	11:30 a.m.
ROOM	Gates 104

Slot error rate: 1/3

Task success: At end, was the correct meeting added to the calendar?



Other Heuristics

➤ Efficiency cost:

- ▶ Total elapsed time for the dialogue in seconds,
- ▶ The number of total turns or of system turns
- ▶ Total number of queries
- ▶ "Turn correction ratio": % of turns that were used to correct errors

➤ Quality cost:

- ▶ Number of ASR rejection prompts.
- ▶ Number of times the user had to barge in



Further Reading

► Speech and Language Processing (3rd ed. draft)

► Chapter 24